

BCAC 0028: SOFT COMPUTING

Objective: This course is designed to provide an introduction to the Soft Computing.

Credits: 04

L-T-P: 4-0-0

Module No.	Content	Teaching Hours
I	Introduction To Neural Networks: Neural Networks Neuron, Nerve Structure And Synapse, Artificial Neuron And Its Model, Activation Functions. Neural Network Architecture: Single Layer And Multilayer Feed Forward Networks, Recurrent Networks. Perception And Convergence Rule. Supervised Learning Network & Unsupervised Learning Network. Back Propagation Networks-I: Perceptron Model, Solution, Single Layer, Multilayer Perception Model. Back Propagation Networks-II: Back Propagation Learning Methods, Effect Of Learning Rule Co-Efficient; Back Propagation Algorithm, Applications.	30
II	Fuzzy Logic Introduction-I: Basic Concepts Of Fuzzy Logic, Fuzzy Sets And Crisp Sets, Fuzzy Set Theory And Operations, Properties Of Fuzzy Sets. Fuzzy Logic Introduction-II: Fuzzy And Crisp Relations, Fuzzy To Crisp Conversion, Membership Functions, Interference In Fuzzy Logic, Fuzzy If-Then Rules, Fuzzyfications & Defuzzificataions. Genetic Algorithm-I: Basic Concepts, Working Principle, Procedures Of GA, Flow Chart Of GA. Genetic Algorithm-II: Genetic Representations, (Encoding), Genetic Operators, Mutation, Generational Cycle.	30

Text Book:

- S. Rajsekaran & G.A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications" Prentice Hall of India, 2003.

Reference Books:

- Anderson, James, "Introduction to Neural Networks", PHI Publication, Delhi, India.
- N.P.Padhy, "Artificial Intelligence and Intelligent Systems" Oxford University Press, USA, 2005.
- Simon Haykin, "Neural Netowrks and Learning Machines "Prentice Hall of India, 2005, Third Edition.

Focus: This course focuses on Employability under CO2, CO3, CO4 and CO5.

Outcome: Upon the completion of this course the student will be able to:

- CO1: Knowledge of soft computing concepts
- CO2: Apply soft computing concepts for practical applications.
- CO3: Design suitable Neural Network for real time problems.
- CO4: Apply fuzzy rules and reasoning to develop decision making and expert systems.
- CO5: Analyze the importance of optimization techniques and genetic programming.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/ PSOs
CO1	PO1,PO2 / PSO3
CO2	PO2/PSO1
CO3	PO3/PSO1,POS2
CO4	PO1,PO4/PSO1,PSO3
CO5	PO3,PO7/PSO1